



USN

--	--	--	--	--	--	--	--	--	--

Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CST113 – ADVANCED DESIGN OF RCC STRUCTURES

(For CST)

Duration: 3 Hours]

[Max. Marks:100

Note: i. Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

ii. Use of IS 456:2000 and SP16 shall be permitted.

iii. Any missing information shall be suitably assumed and duly noted appropriately on the answer script.

iv. Coefficient table shall be supplied for curved beam design.

Unit – I

- | | | Marks | BT* | CO* | PO* |
|----|---|-------|-----|-----|-----|
| 1. | a) With reference to a RCC slab, explain what is meant by Yield Lines?
Enumerate the factors that affect the shape of a Yield line in RCC slabs? | 5 | L2 | 1 | 1,2 |
| | b) Enumerate and discuss in detail, the assumptions and properties of Yield lines in slabs as considered in Yield line theory? Draw neat sketches to indicate pattern of Yield lines in the following cases of slabs?
i. Square Slab with all four edges continuous.
ii. Square Slab with all four edges discontinuous.
iii. Rectangular Slab with all four edges discontinuous.
iv. Circular Slab with edges discontinuous.
v. Triangular Slab with all three edges discontinuous. | 15 | L2 | 1 | 1,2 |
| 2. | Given the following data, design a flat slab without drops or column head using Direct Design Method for flexure in one direction only. The calculations for dimensioning of flat slab and calculation of design moments for Interior panel only is to be made. Make use of simple sketches wherever necessary.
i. Center to Center distance of columns = 6.5m x 6.5m
ii. Size of columns = 500mm x 500mm
iii. Height of Column = 4.5 m
iv. Grade of Concrete = M25
v. Yield Strength of Reinforcement Steel = 500 MPa
vi. Density of Concrete = 25 kN/m ³
vii. Live Load = 3.0 kN/m ²
viii. Floor Finish = 1.5 kN/m ² | 20 | L3 | 1 | 1,6 |
| 3. | Given the following data, proportion and analyze grid slab using the moment coefficients provided in IS 456:2000 for solid slabs?
i. Dimension of the slab – 24m x 30m
ii. Live Load – 3.0 kN/m ²
iii. Spacing of Ribs – 1.5m
iv. Floor Finish – 1.5 kN/m ²
v. Concrete Grade M20
vi. Reinforcement Steel Grade Fe500
vii. End condition – All four edges discontinuous | 20 | L3 | 2 | 2,6 |



Unit – II

4. Given the following data, design a square bunker wall and hopper bottom. For clarity, use simple sketches wherever necessary.
- Plan size = Inner to inner size is 3m x 3m
 - Height of Bunker = 3.5 m
 - Height of Hopper Bottom = 1.0m
 - Plan size of opening of the Hopper Bottom = 0.45 m
 - Angle of surcharge (α) = Angle of Repose/Internal Friction = 30°
 - Density of material stored(γ)= 9 kN/m²
 - Grade of Concrete = M20
 - Yield Strength of Reinforcement Steel = 500 MPa
 - Density of Concrete = 25 kN/m³
- 20 L3 3 3,6
5. Given the following data, calculate the horizontal pressure intensities developed at every 5 m intervals in a Cement Silo using Janssen's or Airy's theory.
- Internal Diameter of Silo = 10 m
 - Height of Cylindrical Portion = 35 m
 - Angle of Repose of Cement (ϕ)= 18 Degrees.
 - Coefficient of friction between concrete and cement (μ')= 0.554
 - Coefficient of friction between filling on filling (μ) = 0.316
 - n =Ratio of Horizontal Pressure intensity (p_h) to Vertical Pressure Intensity(p_v) = 0.528
 - Density of material stored(γ)= 9 kN/m²
 - Density of Cement = 25 kN/m³
- Further design the silo wall for pressure at depth of 35m and wall of the hopper bottom. Use the following additional data.
- Diameter of opening of hopper bottom slab = 1.0 m
 - Slope of hopper bottom slab (θ) = 45 degrees
 - Grade of Concrete = M20
 - Yield Strength of Reinforcement Steel = 500 MPa
 - Lining thickness = 30mm
- 20 L3 3 3,6
6. Consider a simply supported RCC beam of two equal spans of 8m each, with a uniform beam size of 300mm x 600mm. If the UDL on the beam (including self-weight of beam) is 25 kN/m, analyze the beam using any suitable method and design Moment Distribution Method and draw the BMD and SFD for the following two cases. With 20% redistribution of support moment at the intermediate support. Design the beam for flexure and shear after redistribution and draw a longitudinal section of the beam indicating the reinforcement details.
- 20 L3 3 3,5

Unit – III

7. Design a semicircular beam supported on 3 columns equally spaced. The centre of the columns is on a curve of diameter 10m. The superimposed load on the beam per meter length is 18kN. Radius of the girder=6m. Depth of the girder=1000mm. Use M20 concrete and Fe500 steel. Draw cross-sections of beam at support, mid-span and at location of maximum torsion indicating reinforcement details.
- 20 L3 4 4,5
8. A circular RC girder for the foundation raft of a water tower has a mean diameter of 10m. The uniformly distributed load transmitted by 8 symmetrically placed columns on the girder being 450 kN/m. The width of the beam is 500mm and the overall depth is 1000mm. using M20 grade concrete and Fe 500 grade steel, design suitable reinforcements in the circular girder and sketch the details of reinforcements at critical sections. Use the parameters $\theta=45$, $K_1=0.0083$, $K_2=0.0041$, $K_3=0.0006$.
- 20 L3 4 4,6

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome